

Siying Liu

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Education

Columbia University | Master of Arts

Sept 2026 - May 2027

Quantitative Method of Social Science

Relevant Coursework: Quantitative Theory, Advanced Machine Learning, Time Series, Panel Data & Forecasting

University of Wisconsin-Madison | Bachelor of Science

Sept 2023 - May 2026

Majors: Economics - Math Emphasis | Certificate: Data Science, Statistics

GPA: 3.43/4

Relevant Coursework: Econometrics, Probability, Statistic, Machine Learning, Data Visualization, Regression Analysis

Work Experience:

DetailPage

Remote

Content Analyst

Feb 2025– Jan 2026

- Applied AI-enabled analytical tools and keyword-performance data to optimize Amazon product listings, improving average search rankings by 150% and increasing traffic.
- Conducted keyword research across 300+ product listings, identifying high-volume and relevant search terms to strengthen consumer reach and engagement.
- Evaluated more than 1,000 AI-generated SEO keyword recommendations based on search volume, product relevance, customer intent, and expected performance.

AfterQuery

Remote

Analyst

April 2025 - May 2025

- Analyzed 20+ public-company 10-K filings to identify revenue drivers, profitability trends, operating risks, and material changes in financial performance.
- Developed 5+ multi-step financial-analysis questions and solutions used to train and improve AI model reasoning on company disclosures.
- Evaluated model responses for numerical accuracy, logical consistency, and appropriate interpretation of financial information, identifying and correcting 50+ analytical issues in the model.

AI Supermarket

New York City, NY

Market Analyst for Lee Kum Kee

July 2024– August 2024

- Engaged directly with more than 200 customers, assessed individual preferences, and delivered personalized product recommendations to strengthen brand engagement.
- Identified key customer demographic groups and adjusted sampling strategies based on customer behavior, product preferences, and popular dishes.
- Resolved 100+ customer questions or concerns by recommending suitable alternative products, supporting customer satisfaction and brand loyalty.

Project

- Heart Disease Risk Prediction Analysis | Language: R
 - Preprocessed and transformed a heart disease dataset, then implemented and compared filter, embedded, and wrapper feature selection methods to evaluate which clinical variables were most informative for prediction.
 - Produced a concise analytical report that interpreted differences across feature selection approaches, connected results to domain-specific conclusions, and documented the workflow for reproducibility.
- Flight Delays During the 2023 Holiday Season | Language: R
 - Analyzed USDOT commercial flight data for non-stop routes departing/arriving during the 2023 holiday travel season.
 - Defined an “on-time arrival” outcome using arrival delay and examined factors linked to punctual arrivals across airlines, airports, dates, scheduled departure times, weather and routes.
 - Built and compared regression models in R to generate practical travel takeaways and separate guidance for reducing cancellation and delay risk.
- Comparative Analysis of Broken Ice Cream Machines by State | Language: R
 - Conducted statistical hypothesis testing to compare the proportion of broken McDonald's ice cream machines in Wisconsin versus other U.S. states.
 - Cleaned and visualized large-scale time-series data; presented results using ggplot2 and interpreted p-values to determine significance.
- Macroeconomic Indicators Analysis: Unemployment and Personal Consumption | Language: STATA
 - Built a regression model examining the relationship between U.S. unemployment and personal consumption expenditures.
 - Interpreted coefficients, tested model assumptions and produced professional summary tables to communicate economic findings.
- Global Environmental Analysis: Pollution vs. Population Growth Rate | Language: Python
 - Used Python (pandas, matplotlib, seaborn) to explore the relationship between global pollution and human population growth rates.
 - Developed data visualizations and conducted trend analysis to understand long-term environmental impacts.

TECHNICAL SKILLS:

- Programming: Python, R, SQL, STATA, Java
- Analytics: Regression, statistical modeling, hypothesis testing, feature selection, time-series analysis, data visualization
- Tools: Excel, PowerPoint, Word, LaTeX
- Languages: English, Mandarin, Cantonese